2 Water is held behind a dam in a hydroelectric power scheme.

(a) State the main form of energy stored in the water behind the dam.

..... [1]

(b) The water is released from the dam and falls a vertical height of 410 m at a rate of 480 kg / s.

(i) Calculate the rate at which energy is transferred by the falling water.

rate of energy transfer = ... [3]

(ii) The power scheme supplies a current of 270 A at a voltage of 6000 V.

Calculate the efficiency of the power scheme.

efficiency = ...% [3]

(c) Hydroelectric energy is a renewable form of energy.

(i) State **one** disadvantage of hydroelectric power schemes.

..... [1]

(ii) State **one** other renewable source of energy.

..... [1]

[Total: 9]

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[Turn over